ADAPTIQ — ADAPTIVE INTELLIGENCE FOR PERSONALIZED LEARNING

[123] THULIR

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Grounding: 100% Derived from 2 Verified Course Document(s)

PART A: 2-MARK QUESTIONS & VERIFIED ANSWERS

Concise Definitions, Key Invariants, Operational Differences & Formulas

Total Questions: 7 • Marks per Question: 2 Marks • Total Marks: 14 Marks

INDEX OF COVERED TOPICS & LEARNING UNITS:

1. System Engineering • 2. Workflow Management • 3. Analytics & Metrics • 4. Information Security • 5. Infrastructure • 6. Human-Computer Interaction • 7. Networking

QUESTION 1 Topic: System Engineering **[2 Marks]**

Q1. Define the primary purpose of the Project Thulir system architecture and its stakeholder interfaces.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 5) — "High-level system design showcasing core modular boundaries and stakeholder interfaces for Project Thulir...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The system architecture establishes foundational modular boundaries and stakeholder interfaces for Project Thulir. It provides a comprehensive architectural layout demonstrating core systemic integration designed for grassroots innovation management.

QUESTION 2 Topic: Workflow Management **[2 Marks]**

Q2. Describe the operational stages involved in the workflow management evaluation flowchart.

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 18) — "Stage-gate evaluation flowchart mapping ideation filtering down to prototyping approval...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The stage-gate evaluation flowchart maps the systematic filtering process of raw inputs. Specifically, it governs the progression from initial ideation filtering all the way down to prototyping approval.

QUESTION 3 Topic: Analytics & Metrics **[2 Marks]**

Q3. What are the core components displayed on the analytics and metrics dashboard mockup?

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 12) — "Dashboard mockup displaying key performance indicators, impact metrics, and feedback aggregation charts...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The dashboard mockup integrates key performance indicators (KPIs), quantitative impact metrics, and feedback aggregation charts. These elements form the metric framework for evaluating project success.

QUESTION 4 Topic: Information Security **[2 Marks]**

Q4. Explain the role-based mechanisms specified in the information security access control matrix.

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 34) — "Access control matrix detailing role hierarchies, permissions, and data encryption standards...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The access control matrix outlines security measures by detailing role hierarchies, specific system permissions, and mandatory data encryption standards. These protocols protect sensitive user data.

QUESTION 5 Topic: Infrastructure **[2 Marks]**

Q5. Identify the core architectural components that make up the cloud deployment topology.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 42) — "Cloud deployment topology illustrating container clusters, load balancers, and regional failovers...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

The cloud deployment topology comprises scalable container clusters, high-availability load balancers, and regional failover mechanisms. Together, they ensure fault tolerance and system reliability.

QUESTION 6 Topic: Human-Computer Interaction **[2 Marks]**

Q6. How do design frameworks enhance rural user accessibility in Human-Computer Interaction for Project Thulir?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Slide 25) — "Design frameworks emphasizing voice-first navigation for rural user accessibility...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Accessibility for rural users is optimized through design frameworks that incorporate voice-first navigation. Additionally, UI wireframes highlight mentor-mentee matching dashboards to streamline user engagement.

QUESTION 7 Topic: Networking **[2 Marks]**

Q7. State the telemetry transmission protocol utilized over constrained cellular links.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: THULIR_AI_SIH2026_Updated_Board_Book_v3.0 (Page 30) — "Telemetry transmission protocols utilizing MQTT over constrained cellular links...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Telemetry transmission over constrained cellular links relies on the MQTT protocol. This lightweight protocol is optimized for low-bandwidth, high-latency remote environments.